Height-p transport for heat-flowed L-functions

An explicit remainder for the exterior input of strip comparisons

Drafted for Edward Baker

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Abstract

For the de Bruijn–Newman heat flow H_t of the Riemann ξ-function and of the completed Dirichlet L-functions ξ_χ, strip comparisons bound the squared height of the highest zero by a differential inequality whose coefficient is $S(x,p,t) = -\operatorname{Im}(H_t'/H_t)(x+ip)$, read at a fixed height p above the zeros. We show that S is half the real part of the logarithmic derivative of a Gaussian average of the completed function, and that recentring this average at the saddle point of the Gamma factor gives an exact transport identity. We bound its remainder explicitly using only Euler-product majorants at real arguments, the digamma function and a partial-summation bound in the critical strip, with no Riemann–Siegel formula, truncated Dirichlet series or zero-free region. Cauchy's estimate then bounds $\partial_p S$. Evaluated in floating point on a grid of times, the bound keeps the exact time-zero floor of S with no loss for every primitive character of conductor at least $1.1\cdot 10^5$ (even) or $4.9\cdot 10^4$ (odd) at $p=\sqrt{5}$, and for ζ beyond the heights of the published computations of Polymath and of Platt and Trudgian. At Hakimoglu-Brown's height $p = 61/20$ its remainder, evaluated in floating point, is 1.7 to 4.4 times smaller than that of his Riemann–Siegel-based estimate. As a cross-check we restart his derivative-kernel comparison from the classical strip. A 34-row table, verified in exact and directed-rounding arithmetic, gives $\Lambda \le 0.18891$ conditionally on the analytic lemmas of his unrefereed preprint. This is not a record: unrefereed preprints already claim stronger bounds from the same starting point.

Draft status. This working manuscript was requested before the formal review of the project's first round. It integrates round-one Notes 01 and 02 [9, 10] and their check programmes [18]. The analytic statements of Sections 2–5 are proved here. Section 6 reports floating-point evaluations of those bounds, not outward-rounded certificates. Proposition 7.5 is conditional on a cited unrefereed preprint, and Section 8 is planning arithmetic. Derivations, programmes and authoring checks used Claude Opus 5.5 (Anthropic). Those checks were internal and same-model; cross-model and specialist human review are pending.

1 Introduction

1.1 The heat flow and the strip comparison

Let Φ be the Riemann kernel and $H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,\mathrm{d}u$, so that $H_0(z) = \frac{1}{8}\xi\big(\frac{1+iz}{2}\big)$ and $\partial_t H_t = -\partial_z^2 H_t$ [15]. De Bruijn showed that H_t has only real zeros for $t \ge \frac{1}{2}$ [1]. Newman showed that there is a constant $\Lambda \le \frac{1}{2}$ such that H_t has only real zeros exactly when $t \ge \Lambda$ [7], and the Riemann hypothesis is the statement $\Lambda \le 0$. Rodgers and Tao proved $\Lambda \ge 0$ [19]. The published upper bounds are 0.22 [15] and 0.2 [14]. Dobner extended the construction to the extended Selberg class, in particular to ξ_χ for primitive Dirichlet characters, and proved $\Lambda_\chi \ge 0$ [3].

A recent line of work follows the highest zero directly. If $x+ih$ is a highest simple zero of H_t and $Q=h^2$, the zero dynamics give

$$Q' \le -2 - \frac{4Q}{p}\,S(x,p,t) + \frac{8Q}{p^2-Q} \qquad (p^2 \ge 5Q), \qquad S(x,p,t) = -\operatorname{Im}\frac{H_t'}{H_t}(x+ip) \tag{1.1}$$

([6, Section 4]; Proposition 2.2 below). Hence a lower bound for S at one height p and every relevant x drives Q down. Hakimoglu-Brown [6] combines (1.1) with three further inputs: Polymath's finite-region computation, an approximation of Riemann–Siegel type on the band $3 \le \operatorname{Im} z \le 6$, and an upper bound for $\partial_p S$ (his Proposition 11), which enters through a strengthened kernel inequality (his Lemma 10 and Theorem 12). He obtains $\Lambda \le 0.13963$ under the hypothesis that ζ has no zero with $\operatorname{Re} s > \frac{7}{8}$.

Every comparison of this kind needs a floor for $S(x,p,t)$ along the flow, uniform over the real parts x where the highest zero can lie. At $t=0$ the floor is exact. There $S(x,p,0)$ is half the real part of F'/F at $\operatorname{Re} s = \frac{1+p}{2} > 1$, which the Euler product bounds below (Lemma 2.1). This paper is about keeping that floor for $t>0$. In the language of the project it belongs to, it supplies the exterior input of a strip comparison: the statement called Lemma A_p in [8]. It does so without passing through a zero-free region near the critical line.

1.2 Results

Throughout, F is ξ or ξ_χ for a primitive character χ of conductor q and parity $\mathfrak{a}$. Let Γ_F be its Gamma factor and $\alpha = \Gamma_F'/\Gamma_F$. Fix $p>1$ and $0<t\le\frac12$, and put $\tau=\sqrt{t/2}$ and $w=\frac{1+p}{2}+\frac{i}{2}x$.

- *Gaussian form and saddle (Section 3).* We have $S(x,p,t) = \frac12 \operatorname{Re}(\mathcal{G}'/\mathcal{G})(w)$, where $\mathcal{G}(s) = \mathbb{E}\,F(s+\tau N)$ for a standard normal variable N. The fixed point $s_* = w + \frac{t}{2}\alpha(s_*)$ exists and is unique (for ζ, in a box D_x at heights $x \ge 40$, for $p \le x$). Moving the line of integration to s_* gives an exact formula for $(\mathcal{G}'/\mathcal{G})(w) - (F'/F)(s_*)$ (Lemmas 3.1–3.3).

- *Explicit remainder (Section 4).* We have $\big|(\mathcal{G}'/\mathcal{G})(w) - (F'/F)(s_*)\big| \le E$, with E given explicitly by majorants at real arguments (Proposition 4.5). Consequently

$$S(x,p,t) \;\ge\; \tfrac14\ell_* - \tfrac12\kappa(\sigma_*) - \tfrac12 E, \qquad \sigma_* = \tfrac{1+p}{2} + \tfrac{t}{4}\ell_*, \quad \ell_* = 2\operatorname{Re}\alpha(s_*), \tag{1.2}$$

  with $\kappa = -\zeta'/\zeta$ (Theorem 4.6). The time-t floor is therefore the time-zero floor at a point moved to the right by $t\ell_*/4$; this is the *height-p transport*.

- *Derivative bound (Section 5).* Cauchy's estimate applied to the remainder bounds $\partial_p S$ (Corollary 5.1).

- *Numerical evaluation (Section 6).* Floating evaluation of E on a grid of 20 times $0.001 \le t \le \frac12$ shows the floor of Lemma 2.1 persisting with no loss. This holds for every primitive character of conductor $q \ge 1.084\cdot 10^5$ (even) or $q \ge 4.892\cdot 10^4$ (odd) at $p=\sqrt5$, at every height. For ζ the constant floor $\frac14\ell_X - \frac12\kappa(\sigma_0)$ persists at every $x \ge X$, for the heights X used by Polymath [15] and by Platt and Trudgian [14]. At $p=61/20$ the floating remainder is smaller than that of [6, Proposition 11] by factors of 1.7 to 4.4.

- *A check from the classical strip (Section 7).* Conditionally on the analytic lemmas of [6], Hakimoglu-Brown's Theorem 12 started from the classical strip $M(0)\le 1$ gives $\Lambda \le 18891/100000$ (Proposition 7.5). The 34-row table is verified in exact and directed-rounding arithmetic.

Table 1 records the status of each statement.

Statement	Status
Lemmas 2.1–4.4, Propositions 2.2 and 4.5, Theorem 4.6, Corollary 5.1	proved here (Lemma 2.1 and Proposition 2.2 after [8, 6])
Section 6: thresholds, floors, comparison with [6]	numerical: floating point, monotonicity in ℓ_* sampled, grid times only
Proposition 7.5: $\Lambda \le 0.18891$	conditional on [6, Lemmas 2, 3, 6, 10, Proposition 11, Theorem 12] and the further inputs listed in Proposition 7.5
Section 8	planning: depends on unproved inputs named there

Table 1: Status of the statements.

1.3 Relation to other work

The exterior inputs of Polymath [15] and Hakimoglu-Brown [6] come from approximate functional equations: a heat-weighted Dirichlet polynomial, its reflection and Riemann–Siegel remainders. The transport identity replaces these at heights $p > 1$ by an exact Gaussian average with a saddle-point recentring. Its main term coincides with Hakimoglu-Brown's. His heat weight $e^{t\log^2 n/4}$ equals $\mathbb{E}\, n^{-\tau N}$, and his point $w + \frac{t}{2}\alpha(w)$ is, up to complex conjugation, the first iterate of the saddle equation (Section 6.6).

Starting a stepped comparison from the classical strip is not new. Gordon [5], Raymond [16] and Planat [11] start from it and claim $\Lambda \le 0.157656619$, $\Lambda < 0.149396$ and, under stated premises, $\Lambda \le 0.148611789321711$. Gomila [4] claims $\Lambda \le 0.1787854$ by Polymath's criterion at a larger height. All four are unrefereed, and we did not read them in full (Section 7.4). Proposition 7.5 is a cross-check: it rests on published numerical inputs and on the analytic lemmas of [6]. For Dirichlet L-functions we know of no unconditional upper bound below $\frac{1}{2}$ for Λ_χ [8]. There the transport identity is the main new input, and Section 8 records what it would give.

2 Setting, the time-zero floor and the comparison

2.1 Conventions

We write $s = \frac{1}{2}(1 + iz)$ with $z = x + iy$. A point $s = \beta + iT$ corresponds to $z = 2T + i(1 - 2\beta)$, so the real part x is twice the ordinate and the line $\operatorname{Re} s = \sigma$ is the line $\operatorname{Im} z = 1 - 2\sigma$.

The family. Let χ be a primitive Dirichlet character modulo $q \ge 3$ with $\chi(-1) = (-1)^{\mathfrak{a}}$, $\mathfrak{a} \in \{0, 1\}$. Put $\mathcal{L} = L(\cdot, \chi)$, $\Gamma_F(s) = \epsilon_\chi (q/\pi)^{s/2}\Gamma(\frac{s+\mathfrak{a}}{2})$ and $F = \xi_\chi = \Gamma_F\mathcal{L}$, where the constant ϵ_χ of modulus one is chosen so that $F(s) = \overline{F(1-\bar{s})}$ [3].

The zeta case. Here $\mathcal{L} = \zeta$, $\Gamma_F(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$, $F = \xi = \Gamma_F\zeta$, and we put $q = 1$ and $\mathfrak{a} = 0$.

In both cases $\alpha = \Gamma_F'/\Gamma_F$ is given by

$$\alpha(s) = \tfrac{1}{2}\log\tfrac{q}{\pi} + \tfrac{1}{2}\psi\big(\tfrac{s+\mathfrak{a}}{2}\big) \quad \text{(family)}, \qquad \alpha(s) = \tfrac{1}{s} + \tfrac{1}{s-1} - \tfrac{1}{2}\log\pi + \tfrac{1}{2}\psi\big(\tfrac{s}{2}\big) \quad \text{(zeta)}, \tag{2.1}$$

with $\psi = \Gamma'/\Gamma$. Let $H_t(z) = \int_{\mathbb{R}} e^{tu^2}\Phi(u)e^{izu}\,\mathrm{d}u$, where $\Phi = \Phi_\chi$ is Dobner's kernel [3] for the family and Polymath's kernel [15] for ζ. Then $\partial_t H_t = -\partial_z^2 H_t$, H_t is real on $\mathbb{R}$, and $H_0(z) = c_H F(\frac{1}{2}(1+iz))$ with a real constant $c_H \ne 0$. (Polymath's H_t, defined by $\int_0^\infty \cos(zu)\,\mathrm{d}u$, is half of ours; constant factors do not change S.) The kernel decays like $\exp(-\frac{\pi}{q}e^{4|u|})$, so each H_t is entire of order at most one.

Let $M(t) = \sup\{(\operatorname{Im} z)^2 : H_t(z) = 0\}$. Every zero of F has $0 < \operatorname{Re} s < 1$, so $M(0) \le 1$. De Bruijn's strip theorem [1] (for the family see [3]) gives $M(t) \le \max(M(t_0) - 2(t - t_0), 0)$ for $t \ge t_0$,

and in particular $\Lambda \le t_0 + \frac{1}{2}M(t_0)$. We call this the *classical strip theorem*. It implies that every zero of H_t, $t \ge 0$, satisfies $|\operatorname{Im} z| \le 1$.

Fix $p > 1$ and put

$$\sigma_0 = \tfrac{1}{2}(1+p), \qquad w = \sigma_0 + \tfrac{i}{2}x, \qquad \tau = \sqrt{t/2},$$

so that $z = x + ip$ corresponds to $s = 1 - \bar{w}$. Let φ_τ be the density of the centred normal law of variance τ^2, and let N be a standard normal variable. For $\sigma > 1$ let

$$\kappa(\sigma) = -\frac{\zeta'}{\zeta}(\sigma) = \sum_{n \ge 2} \Lambda_{\mathrm{vM}}(n) n^{-\sigma}, \qquad K_2 = -\kappa', \qquad K_3 = \kappa'',$$

with Λ_{vM} the von Mangoldt function. These are positive, κ and K_2 decrease, and $\log\zeta$ and κ are convex.

2.2 The time-zero floor

Lemma 2.1 (time-zero floor; after [8, Lemma 3.4])**.** *Let $p > 1$. For the family, at every real x,*

$$S(x,p,0) \;\ge\; C_{q,\mathfrak{a}}(p) := \tfrac{1}{4}\Big(\log\tfrac{q}{\pi} + \psi\big(\tfrac{\sigma_0+\mathfrak{a}}{2}\big)\Big) - \tfrac{1}{2}\kappa(\sigma_0). \tag{2.2}$$

For ζ, at every $x > 0$, $S(x,p,0) \ge \frac{1}{2}\operatorname{Re}\alpha(w) - \frac{1}{2}\kappa(\sigma_0)$, and $\frac{1}{2}\operatorname{Re}\alpha(w) \ge \frac{1}{4}\log\frac{x}{4\pi} - \frac{1}{4}c(\frac{w}{2})$ with c as in Lemma 4.3.

Proof. By Lemma 3.1 at $t = 0$, $S(x,p,0) = \frac{1}{2}\operatorname{Re}(F'/F)(w)$, and $F'/F = \alpha + \mathcal{L}'/\mathcal{L}$. Since $\mathcal{L}'/\mathcal{L}(s) = -\sum \chi(n)\Lambda_{\mathrm{vM}}(n)n^{-s}$, $|(\mathcal{L}'/\mathcal{L})(w)| \le \kappa(\sigma_0)$. For $\operatorname{Re} z > 0$, $\psi(z) = -\gamma_E + \sum_{n\ge0}\big(\frac{1}{n+1} - \frac{1}{n+z}\big)$ and $\operatorname{Re}\frac{1}{n+z} \le \frac{1}{n+\operatorname{Re} z}$, so $\operatorname{Re}\psi(z) \ge \psi(\operatorname{Re} z)$. This gives the family bound. For ζ, $\operatorname{Re}(\frac{1}{w} + \frac{1}{w-1}) > 0$ and, by Lemma 4.3 with $z = w/2$, $\operatorname{Re}\psi(\frac{w}{2}) \ge \log\frac{|w|}{2} - c(\frac{w}{2})$. Hence $\operatorname{Re}\alpha(w) \ge \frac{1}{2}\log\frac{|w|}{2\pi} - \frac{1}{2}c(\frac{w}{2})$, and $|w| \ge \frac{x}{2}$. □

2.3 The paired-zero inequality

For $p, h > 0$, $d \in \mathbb{R}$ and $0 \le v$, let

$$K_p(d,v) = \frac{p-v}{d^2 + (p-v)^2} + \frac{p+v}{d^2 + (p+v)^2} = \frac{2p\,(d^2 + p^2 - v^2)}{[d^2 + (p-v)^2]\,[d^2 + (p+v)^2]}.$$

For a zero $z_k = x_k + iy_k$ with $y_k \ge 0$ write $d_k = x - x_k$, $v_k = y_k$, and give it the weight $\varpi_k = 1$ if $y_k > 0$ and $\varpi_k = \frac{1}{2}$ if $y_k = 0$. Zeros are counted with multiplicity.

Proposition 2.2 (paired-zero inequality; after [6, Section 4] and [8, Proposition 3.3])**.** *Let $p > 1$ and $t > 0$.*

(i) $S(x,p,t) = \sum_{y_k \ge 0} \varpi_k K_p(d_k, v_k)$*, summed over the zeros of H_t; the series converges absolutely.*

(ii) Let $z_j = x + ih$, $h > 0$, be a simple zero of H_t. Its trajectory is differentiable in t, and $h' = -\frac{1}{h} - 2\sum{}' \varpi_k K_h(d_k, v_k)$, where $\sum'$ omits z_j and $\bar{z}_j$.

(iii) If moreover no zero of H_t has imaginary part larger than h, and $p^2 \ge 5h^2$, then $Q = h^2$ satisfies (1.1).

Proof. Since H_t has order at most one and is real on $\mathbb{R}$, its zeros are closed under conjugation and $H_t(z) = z^m e^{a+bz}\prod_k(1-z/z_k)e^{z/z_k}$ with b real. Then $(H_t'/H_t)(z) = \frac{m}{z}+b+\sum_k\big(\frac{1}{z-z_k}+\frac{1}{z_k}\big)$, absolutely convergent. Group each nonreal z_k with $\bar z_k$. The terms $\frac{1}{z_k}+\frac{1}{\bar z_k}$ are real, and $\operatorname{Im}\big(\frac{1}{z-z_k}+\frac{1}{z-\bar z_k}\big) = -K_p(d_k,v_k)$ at $z = x+ip$. A real zero contributes $\operatorname{Im}\frac{1}{z-x_k} = -\frac12 K_p(d_k,0)$. This proves (i). For (ii), the implicit function theorem gives a differentiable trajectory with $z_j' = -\partial_t H_t/\partial_z H_t = \partial_z^2 H_t/\partial_z H_t$ at z_j. Writing $H_t = (z-z_j)g$, this equals $2(g'/g)(z_j) = 2\big[\frac{m}{z_j}+b+\frac{1}{z_j}+\sum_{k\neq j}\big(\frac{1}{z_j-z_k}+\frac{1}{z_k}\big)\big]$. In the imaginary part the terms $\frac{1}{z_j}$ and $\frac{1}{\bar z_j}$ combine to a real number, the conjugate zero contributes $\operatorname{Im}\frac{2}{z_j-\bar z_j} = -\frac{1}{h}$, each other pair contributes $-2K_h(d_k,v_k)$, and each real zero $-K_h(d_k,0)$, as in (i).

For (iii) we use the kernel inequality $K_h(d,v) \ge \frac{h}{p}K_p(d,v)$ for $0\le v\le h$, $(d,v)\neq(0,h)$ and $p^2\ge 5h^2$. With $D=d^2$, $A=h^2$, $B=p^2$, $r=v^2\le A$, $w_A = D+A-r$ and $w_B = D+B-r$, the denominators of K_h and K_p are w_A^2+4Dr and w_B^2+4Dr, and

$$\frac{K_h}{h}-\frac{K_p}{p} = \frac{2(B-A)\big(w_Aw_B-4Dr\big)}{(w_A^2+4Dr)(w_B^2+4Dr)}, \qquad w_Aw_B-4Dr = D^2+(A+B-6r)D+(A-r)(B-r),$$

which is nonnegative because $A+B-6r\ge B-5A\ge 0$. Every other zero has $v_k\le h$, so by (i) and (ii)

$$Q' = 2hh' = -2-4h{\sum}'\varpi_kK_h \le -2-\frac{4Q}{p}\Big(S(x,p,t)-K_p(0,h)\Big), \qquad K_p(0,h) = \frac{2p}{p^2-Q},$$

which is (1.1). □

Hakimoglu-Brown notes that the kernel inequality also appears in Gordon's preprint [5, Section 5]. Lemma 7.2 below strengthens it by a term in ∂_pK_p, which Theorem 7.4 pays for with an upper bound for ∂_pS.

2.4 The exterior input

A comparison built on (1.1) needs, at one fixed height p,

$$\textbf{(E}_p\textbf{)} \qquad S(x,p,t) \;\ge\; C(t) \qquad (0\le t\le t_*,\ x \text{ in the region where the highest zero can lie}). \tag{2.3}$$

For the family at large conductor the region is the whole line. For ζ and for small conductors it is $|x|\ge X$, and the rest is covered by a finite verification and a barrier, as in [6, Lemma 2]. In [8], (2.3) is called Lemma A_p. The comparison also needs uniform confinement of the nonreal zeros at positive times ([8], item H12) and a first-contact argument; for ζ, [6] supplies these through [15, Theorem 1.5] and its Lemma 2. The theorems of this paper concern (2.3) only. Statement (2.3) lives at height $p>1$, where the Dirichlet series converges absolutely: it bounds a Poisson-smoothed density of zeros from below, and uses no zero-free region and no derivative of H_t beyond the first.

3 The transport identity

Lemma 3.1 (Gaussian form)**.** *For $t\ge 0$ and $z\in\mathbb{C}$, $H_t(z) = \mathbb{E}\,H_0(z+i\sqrt{2t}\,N)$. With $\mathcal{G}(s) = \mathbb{E}\,F(s+\tau N)$ one has $H_t(z) = c_H\,\mathcal{G}(\frac12(1+iz))$, $\mathcal{G}(s) = \overline{\mathcal{G}(1-\bar s)}$, and, for $p>1$,*

$$S(x,p,t) = \tfrac12\operatorname{Re}\frac{\mathcal{G}'}{\mathcal{G}}(w).$$

Proof. By Fubini, which the decay of Φ permits, $\mathbb{E}\,H_0(z+i\beta N) = \int \Phi(u)e^{izu}\mathbb{E}\,e^{-\beta N u}\,\mathrm{d}u = \int e^{\beta^2u^2/2}\Phi(u)e^{izu}\,\mathrm{d}u$, and $\beta=\sqrt{2t}$ gives H_t. Substituting $H_0(z)=c_HF(\frac12(1+iz))$ and using $N\overset{d}{=}-N$ gives $H_t(z)=c_H\mathcal{G}(\frac12(1+iz))$ with $\tau=\beta/2$, and the functional equation of F passes to $\mathcal{G}$. Then $H_t'(z)=\frac{i}{2}c_H\mathcal{G}'(\frac12(1+iz))$. At $z=x+ip$, $\frac12(1+iz)=1-\bar w$, and differentiating $\mathcal{G}(s)=\overline{\mathcal{G}(1-\bar s)}$ gives $(\mathcal{G}'/\mathcal{G})(1-\bar w)=-\overline{(\mathcal{G}'/\mathcal{G})(w)}$. So $S=-\operatorname{Im}\big(\frac{i}{2}(\mathcal{G}'/\mathcal{G})(1-\bar w)\big)=\frac12\operatorname{Re}(\mathcal{G}'/\mathcal{G})(w)$. Here $\mathcal{G}(w)\neq0$, because every zero of H_t has $|\operatorname{Im} z|\le 1<p$. □

Lemma 3.2 (the Gamma saddle)**.** Family. *Let $p>1$, $0<t\le\frac12$ and $\ell_0=\log\frac{q}{\pi}+\psi(\frac{\sigma_0+\mathfrak{a}}{2})>0$. The map $\mu\mapsto\tau^2\alpha(w+\mu)$ sends the closed half-plane $\{\operatorname{Re}\mu\ge0\}$ into itself and is a contraction there, so it has a unique fixed point μ_*. Put $s_*=w+\mu_*$, $\ell_*=2\operatorname{Re}\alpha(s_*)$ and $\sigma_*=\operatorname{Re}s_*$. Then*

$$\sigma_*=\sigma_0+\tfrac12\tau^2\ell_*=\sigma_0+\tfrac{t}{4}\ell_*,\qquad \ell_*\ge\ell_*^{(0)}\ge\ell_0,$$

where $\ell_*^{(0)}=2\alpha(\sigma^{(0)})$ and $\sigma^{(0)}$ is the unique real solution of $\sigma=\sigma_0+\tau^2\alpha(\sigma)$, the saddle at $x=0$.

Zeta. *For $x\ge40$ and $1<p\le x$ the same map has a unique fixed point in $D_x=\{\mu:0\le\operatorname{Re}\mu\le R_x,\ |\operatorname{Im}\mu|\le\frac12\}$, $R_x=\frac18\log x+1$. There $\operatorname{Im}s_*\ge\frac{x}{2}-\frac12$ and $\ell_*\ge\log\big((\frac{x}{2}-\frac12)/2\pi\big)-2c(\frac12 s_*)$, with c as in Lemma 4.3. (S is even in x for ζ.)*

Proof. Family. For $\operatorname{Re}\mu\ge0$, $\operatorname{Re}(w+\mu)\ge\sigma_0$. As in the proof of Lemma 2.1, $\operatorname{Re}\psi(z)\ge\psi(\operatorname{Re}z)$, and termwise from $\psi'(z)=\sum_{n\ge0}(n+z)^{-2}$, $|\psi'(z)|\le\psi'(\operatorname{Re}z)$. So $\operatorname{Re}\tau^2\alpha(w+\mu)\ge\frac12\tau^2\ell_0>0$ and $\big|\frac{\mathrm{d}}{\mathrm{d}\mu}\tau^2\alpha(w+\mu)\big|\le\frac14\tau^2\psi'(\frac{\sigma_0+\mathfrak{a}}{2})\le\frac{1}{16}\psi'(\frac12)=\frac{\pi^2}{32}<0.31$ (below 0.15 when $p\ge\sqrt5$). Banach's theorem gives μ_*, and $\operatorname{Re}\mu_*=\tau^2\operatorname{Re}\alpha(s_*)=\frac12\tau^2\ell_*$. Next, $\ell_*\ge2\alpha(\sigma_*)$ and $\sigma_*\ge\sigma_0+\tau^2\alpha(\sigma_*)$. The function $f(\sigma)=\sigma-\sigma_0-\tau^2\alpha(\sigma)$ is strictly increasing ($\tau^2\alpha'<1$) with $f(\sigma^{(0)})=0\le f(\sigma_*)$, so $\sigma_*\ge\sigma^{(0)}\ge\sigma_0$, and α increases on the real axis.

Zeta. For $\mu\in D_x$, $s=w+\mu$ has $\operatorname{Re}s\in[\sigma_0,\sigma_0+R_x]$ and $\operatorname{Im}s\ge\frac{x}{2}-\frac12\ge19.5$. There $|\alpha'(s)|\le2/\operatorname{Im}s\le0.11$ by Lemma 4.2, so the map is a contraction. With $z=s/2$, $\operatorname{Im}\psi(z)=\sum_{n\ge0}\operatorname{Im}z/|z+n|^2\in(0,\frac{\pi}{2}+\frac{1}{2\operatorname{Re}z}]$ and $|\operatorname{Im}(\frac1s+\frac{1}{s-1})|\le2/\operatorname{Im}s$, so $|\operatorname{Im}\tau^2\alpha(s)|\le\frac14(\frac{\pi}{4}+\frac12+0.11)<\frac12$. Lemma 4.3 and $\operatorname{Re}(\frac1s+\frac{1}{s-1})>0$ give $\frac12\log\frac{|s|}{2\pi}-c(z)\le\operatorname{Re}\alpha(s)\le\frac12\log\frac{|s|}{2\pi}+\frac{1}{12\operatorname{Re}z|z|}+\frac{2}{\operatorname{Im}s}$, the upper bound from $\operatorname{Re}\psi(z)\le\log|z|+|I(z)|$ in the notation of that lemma. The lower bound is positive. Since $p\le x$, $|s|\le\operatorname{Re}s+\operatorname{Im}s\le x+2+\frac18\log x\le2x$ on D_x, so $\tau^2\operatorname{Re}\alpha(s)\le\frac18\log\frac{x}{\pi}+0.03<R_x$. So D_x is mapped into itself. The bound on ℓ_* is the lower bound at s_*. □

Lemma 3.3 (the shifted representation)**.** *Let $\Theta(v)=F(s_*+v)e^{-\alpha(s_*)v}/F(s_*)$, an entire function of v of order one with $\Theta(0)=1$, and $c_1=(\mathcal{L}'/\mathcal{L})(s_*)$. (Since $\operatorname{Re}s_*>1$, $F(s_*)\neq0$.) Then*

$$\frac{\mathcal{G}'}{\mathcal{G}}(w)-\frac{F'}{F}(s_*)=\frac{\int_{\mathbb{R}}(\Theta'-c_1\Theta)\varphi_\tau}{\int_{\mathbb{R}}\Theta\varphi_\tau},\qquad \int_{\mathbb{R}}(\Theta'-c_1\Theta)\varphi_\tau=\int_{\mathbb{R}}\Big(\frac{v}{\tau^2}-c_1\Big)\Theta\varphi_\tau.$$

Proof. The function $h\mapsto F^{(k)}(w+h)\varphi_\tau(h)$ is entire, and on $\operatorname{Re}h=\pm R$, between 0 and $\operatorname{Im}\mu_*$, it is $O(e^{R^{1+\epsilon}-R^2/2\tau^2})$. So the line of integration in $\mathcal{G}^{(k)}(w)=\int F^{(k)}(w+h)\varphi_\tau(h)\,\mathrm{d}h$ moves to $\mu_*+\mathbb{R}$. Since $\mu_*/\tau^2=\alpha(s_*)$, $\varphi_\tau(\mu_*+v)=c_0\varphi_\tau(v)e^{-\alpha(s_*)v}$ with $c_0=e^{-\mu_*^2/2\tau^2}$. Hence $\mathcal{G}(w)=c_0F(s_*)\int\Theta\varphi_\tau$ and $\mathcal{G}'(w)=c_0F(s_*)\int(\Theta'+\alpha(s_*)\Theta)\varphi_\tau$, because $F'(s_*+v)e^{-\alpha(s_*)v}/F(s_*)=\Theta'(v)+\alpha(s_*)\Theta(v)$. Subtract $(F'/F)(s_*)=\alpha(s_*)+c_1$. The second identity is Gaussian integration by parts, $\varphi_\tau'=-(v/\tau^2)\varphi_\tau$, valid because Θ has order one. □

The question is therefore how far the complex weight $\Theta\varphi_\tau$ is from the pure tilt $e^{c_1v}\varphi_\tau$, for which the right side vanishes. Where $\operatorname{Re}(s_*+v)>1$ we have $\Theta=e^{c_1v+\omega(v)}$ with

$$\omega(v)=\big[\log\mathcal{L}(s_*+v)-\log\mathcal{L}(s_*)-c_1v\big]+\big[\log\Gamma_F(s_*+v)-\log\Gamma_F(s_*)-\alpha(s_*)v\big],$$

and $\Theta'-c_1\Theta=\Delta\Theta$ with $\Delta(v)=(F'/F)(s_*+v)-(F'/F)(s_*)$.

4 An explicit remainder

For a function f of a real variable write $[\![f]\!]_v = f(\sigma_* + v) - f(\sigma_*) - v f'(\sigma_*)$ for its Taylor remainder at σ_*.

Lemma 4.1 (Euler-product majorants)**.** *For* $\operatorname{Re} s = \sigma > 1$ *and* $j \geq 0$, $|(\mathcal{L}'/\mathcal{L})^{(j)}(s)| \leq (-1)^j \kappa^{(j)}(\sigma)$, $|\log \mathcal{L}(s)| \leq \log \zeta(\sigma)$ *and* $\zeta(2\sigma)/\zeta(\sigma) \leq |\mathcal{L}(s)| \leq \zeta(\sigma)$. *Hence, for real* v *with* $\sigma_* + \min(v, 0) > 1$,

$$\begin{aligned} |(\mathcal{L}'/\mathcal{L})(s_* + v) - c_1| &\leq |\kappa(\sigma_* + v) - \kappa(\sigma_*)|, \\ |(\mathcal{L}'/\mathcal{L})(s_* + v) - c_1 - v(\mathcal{L}'/\mathcal{L})'(s_*)| &\leq [\![\kappa]\!]_v, \\ |\log \mathcal{L}(s_* + v) - \log \mathcal{L}(s_*)| &\leq |\log \zeta(\sigma_* + v) - \log \zeta(\sigma_*)|, \\ |\log \mathcal{L}(s_* + v) - \log \mathcal{L}(s_*) - c_1 v| &\leq [\![\log \zeta]\!]_v. \end{aligned}$$

Proof. $\mathcal{L}'/\mathcal{L} = -\sum \chi(n)\Lambda_{\mathrm{vM}}(n) n^{-s}$ and $\log \mathcal{L} = \sum \chi(n)\Lambda_{\mathrm{vM}}(n)(\log n)^{-1} n^{-s}$ with $|\chi(n)| \leq 1$; the bounds on $|\mathcal{L}|$ follow from the Euler product. Each difference is an integral along the horizontal segment from s_* to $s_* + v$ of a derivative bounded by a majorant of fixed sign, whose integral is the corresponding difference or Taylor remainder of κ or $\log \zeta$; both are convex. □

Lemma 4.2 (Gamma majorants)**.** Family. *For* $\operatorname{Re} z > 0$, $|\psi'(z)| \leq \psi'(\operatorname{Re} z)$ *and* $|\psi''(z)| \leq -\psi''(\operatorname{Re} z)$. *Put* $\gamma_\Gamma(\sigma) = -\frac{1}{2}\psi(\frac{\sigma+\mathfrak{a}}{2})$ *and* $\lambda_\Gamma(\sigma) = \log \Gamma(\frac{\sigma+\mathfrak{a}}{2})$. *For real* v *with* $\sigma_* + \min(v, 0) > -\mathfrak{a}$,

$$|\alpha(s_* + v) - \alpha(s_*)| \leq |\gamma_\Gamma(\sigma_* + v) - \gamma_\Gamma(\sigma_*)|, \qquad |\alpha(s_* + v) - \alpha(s_*) - v\alpha'(s_*)| \leq [\![\gamma_\Gamma]\!]_v,$$
$$|\log \Gamma_F(s_* + v) - \log \Gamma_F(s_*) - \alpha(s_*)v| \leq [\![\lambda_\Gamma]\!]_v =: \vartheta(v).$$

Zeta. *If* $|\operatorname{Im} s| \geq T \geq 10$, *at any real part,* $|\alpha'(s)| \leq 2/T$ *and* $|\alpha''(s)| \leq 3/T^2$. *Put* $\vartheta(v) = v^2/T$.

Proof. The family bounds are integrals along the segment, as in Lemma 4.1, since $|\alpha'| \leq -\gamma_\Gamma'$, $|\alpha''| \leq \gamma_\Gamma''$ and $\lambda_\Gamma'' = -\gamma_\Gamma'$. For ζ, with $z = s/2 = a + ib$, the summands of $\sum_{n\geq 0}|z+n|^{-2}$ and $\sum_{n\geq 0}|z+n|^{-3}$ are unimodal in n, so each sum is at most the integral plus the largest term: $\sum_{n\geq 0}|z+n|^{-2} \leq b^{-2} + \pi/|b|$ and $\sum_{n\geq 0}|z+n|^{-3} \leq |b|^{-3} + 2b^{-2}$, with $|b| \geq T/2$. Hence $\frac{1}{4}|\psi'(z)| \leq \frac{\pi}{2T} + \frac{1}{T^2}$ and $\frac{1}{8}|\psi''(z)| \leq \frac{2}{T^3} + \frac{2}{T^2}$. Adding $|s|^{-2} + |s-1|^{-2} \leq 2T^{-2}$ and $2|s|^{-3} + 2|s-1|^{-3} \leq 4T^{-3}$ gives $|\alpha'| \leq \frac{\pi}{2T} + \frac{3}{T^2} \leq \frac{2}{T}$ and $|\alpha''| \leq \frac{2}{T^2} + \frac{6}{T^3} \leq \frac{3}{T^2}$ for $T \geq 10$. □

Lemma 4.3 (a Stirling inequality)**.** *For* $\operatorname{Re} z > 0$,

$$\log|z| \;\leq\; \operatorname{Re}\psi(z) + c(z), \qquad c(z) = \frac{\operatorname{Re} z}{2|z|^2} + \frac{1}{12 \operatorname{Re} z\, |z|}.$$

Proof. Binet's formula [2, Section 5.9(ii)] gives $\psi(z) = \log z - \frac{1}{2z} - I(z)$ with $I(z) = \int_0^\infty \beta(u) e^{-zu}\,\mathrm{d}u$ and $\beta(u) = \frac{1}{e^u - 1} - \frac{1}{u} + \frac{1}{2}$. Since $\beta(0) = 0$, $I(z) = z^{-1}\int_0^\infty \beta'(u) e^{-zu}\,\mathrm{d}u$, and $\beta'(u) = \frac{1}{4} f(u/2)$ with $f(v) = v^{-2} - \sinh^{-2} v$. Here $f > 0$ because $\sinh v > v$, and $f \leq \frac{1}{3}$: for $v \geq \sqrt{3}$ because $v^{-2} \leq \frac{1}{3}$, and for $v < \sqrt{3}$ because $\sinh^2 v\,(3 - v^2) = \sum_{n\geq 1}(3a_n - a_{n-1})v^{2n}$ with $a_n = 2^{2n-1}/(2n)!$ and $a_0 = 0$, whose coefficients are 3, 0 and then $a_{n-1}\big(\frac{12}{2n(2n-1)} - 1\big) < 0$, so that $\sinh^2 v\,(3 - v^2) \leq 3v^2$. Hence $0 \leq \beta' \leq \frac{1}{12}$, $|I(z)| \leq 1/(12 \operatorname{Re} z\, |z|)$, and taking real parts gives the claim. □

For the family put $c_* = \frac{1}{2x_*} + \frac{1}{12x_*^2}$ with $x_* = \frac{1}{2}(\sigma_* + \mathfrak{a})$. It bounds $c(z_*)$, $z_* = \frac{1}{2}(s_* + \mathfrak{a})$, at every height. Since $\ell_* = \log \frac{q}{\pi} + \operatorname{Re}\psi(z_*)$, Lemma 4.3 gives $\log|z_*| \leq \ell_* - \log \frac{q}{\pi} + c_*$, so for $0 < u \leq \sigma_*$

$$q\,|u + i \operatorname{Im} s_*| \;\leq\; 2q|z_*| \;\leq\; \mathcal{Q}_* := 2\pi e^{\ell_* + c_*}. \tag{4.1}$$

For ζ the same $\mathcal{Q}_*$, with $c_* = c(\frac{1}{2}s_*)$, bounds $|u + i \operatorname{Im} s_*|$.

Lemma 4.4 (partial summation in the strip)**.** *For $\sigma > 0$, $s = \sigma + iT$ and every integer $Y \geq 1$,*

$$|L(s,\chi)| \leq 1 + \int_1^Y u^{-\sigma}\,\mathrm{d}u + \frac{q|s|Y^{-\sigma}}{2\sigma} \qquad (\chi \text{ primitive modulo } q \geq 3),$$
$$|\zeta(s)| \leq 1 + \int_1^Y u^{-\sigma}\,\mathrm{d}u + \frac{Y^{1-\sigma}}{|s-1|} + \frac{1}{2}Y^{-\sigma} + \frac{|s|Y^{-\sigma}}{2\sigma}.$$

Proof. $L(s,\chi) = \sum_{n\leq Y}\chi(n)n^{-s} + s\int_Y^\infty (A(u)-A(Y))u^{-s-1}\,\mathrm{d}u$ with $A(u) = \sum_{n\leq u}\chi(n)$. A sum of χ over consecutive integers equals a sum over $r < q$ of them, which is also minus the sum over the other $q-r$ of a period, so $|A(u)-A(Y)| \leq q/2$. For ζ, $\zeta(s) = \sum_{n\leq Y} n^{-s} + \frac{Y^{1-s}}{s-1} - \frac{1}{2}Y^{-s} - s\int_Y^\infty (\{u\}-\frac{1}{2})u^{-s-1}\,\mathrm{d}u$. Both identities hold for $\sigma > 1$ and extend to $\sigma > 0$. □

In the evaluation we take Y near $\mathcal{Q}_*/2$, and for ζ we use $|s-1| \geq |\operatorname{Im} s| \geq T$. Lemma 4.4 is weaker than the convexity bound by about $(q|s|)^{(1-\sigma)/2}$. An explicit convexity bound would improve the small-conductor thresholds of Section 6; we do not use one, so no unread import enters.

Proposition 4.5 (the remainder)**.** *Assume the hypotheses of Lemma 3.2, and for ζ let $10 \leq T \leq \operatorname{Im} s_*$ (for instance $T = \frac{x}{2} - \frac{1}{2}$). Let $t \in (0, \frac{1}{2}]$ and $V \in (0, \sigma_* - 1)$. Write*

$$k_* = K_2(\sigma_*) + A_1(\sigma_*), \quad A_1(\sigma) = \tfrac{1}{4}\psi'\big(\tfrac{\sigma+\mathfrak{a}}{2}\big) \qquad (\textit{zeta: } k_* = K_2(\sigma_*) + 2/T),$$

$\kappa_* = \kappa(\sigma_*)$, and $\mathfrak{M} = \kappa + \gamma_\Gamma$ (zeta: $\mathfrak{M} = \kappa$). For $v \geq -V$ define

$$r(v) = [\![\mathfrak{M}]\!]_v \quad \big(\textit{zeta: } [\![\kappa]\!]_v + \tfrac{3}{2}v^2/T^2\big), \qquad \omega_m(v) = [\![\log\zeta]\!]_v + \vartheta(v),$$
$$g(v) = \exp\big(|\log\zeta(\sigma_*+v) - \log\zeta(\sigma_*)| + \vartheta(v)\big),$$

and for $v < -V$, with $\sigma = \sigma_* + v$ and $z_\zeta = \zeta(\sigma_*)/\zeta(2\sigma_*)$,

$$g_L(v) = \begin{cases} z_\zeta\,\mathcal{Z}(\sigma)\,e^{\vartheta(v)}, & \frac{1}{2} \leq \sigma < \sigma_* - V, \\ z_\zeta\,\mathcal{Z}(1-\sigma)\exp\big(\frac{1}{2}\ell_*(1-2\sigma) + \vartheta(1-\sigma-\sigma_*)\big), & \sigma < \frac{1}{2}. \end{cases}$$

Here $\mathcal{Z}(u) = \zeta(u)$ for $u \geq \sigma_*$; for $1 < u < \sigma_*$ it is the smaller of $\zeta(u)$ and the bound of Lemma 4.4 at real part u, for a fixed integer $Y \geq 1$, with $q|s|$ (zeta: $|s|$) replaced by $\mathcal{Q}_*$ and, for ζ, $|s-1|$ by T; and for $u \leq 1$ it is that bound alone. Put

$$M_\rho = \int_{-V}^\infty r\,g\,\varphi_\tau, \qquad B_V = g(-V)\varphi_\tau(V), \qquad P_1 = \int_{-V}^\infty e^{\kappa_*|v|}\big(e^{\omega_m}-1\big)\varphi_\tau,$$
$$D_L = \int_{-\infty}^{-V} g_L\varphi_\tau, \qquad N_L = \int_{-\infty}^{-V} g_L\Big(\frac{|v|}{\tau^2} + \kappa_*\Big)\varphi_\tau, \qquad P_2 = \int_{-\infty}^{-V}\big(g_L + e^{\kappa_*|v|}\big)\varphi_\tau,$$

and $d_0 = e^{-\tau^2\kappa_*^2/2} - P_1 - P_2$. If $\tau^2 k_* < 1$ and $d_0 > 0$, then

$$\Big|\frac{\mathcal{G}'}{\mathcal{G}}(w) - \frac{F'}{F}(s_*)\Big| \;\leq\; E := \frac{\tau^2 k_*\kappa_*}{1-\tau^2 k_*} + \frac{M_\rho + B_V + \tau^2 k_*\kappa_* D_L + (1+\tau^2 k_*)N_L}{(1-\tau^2 k_*)\,d_0}. \tag{4.2}$$

Proof. Write $\mathcal{N}$ and $\mathcal{D}$ for the numerator and denominator of Lemma 3.3. On $[-V,\infty)$, Lemmas 4.1 and 4.2 give $|\Theta| \leq g$ and $|\omega| \leq \omega_m$. With $\Delta_1 = \Delta'(0) = (F'/F)'(s_*)$ they also give $|\Delta_1| \leq k_*$ and $|\rho| \leq r$ for $\rho(v) = \Delta(v) - \Delta_1 v$.

For $v < -V$ we have $|\Theta(v)| \leq g_L(v)$. When $\sigma \geq \frac{1}{2}$, $|\Theta| = |\mathcal{L}(\sigma + i\operatorname{Im} s_*)/\mathcal{L}(s_*)|\exp\operatorname{Re}\big[\log\Gamma_F(s_*+v) - \log\Gamma_F(s_*) - \alpha(s_*)v\big]$, and Lemmas 4.1–4.4 and (4.1) apply, with $|\mathcal{L}(s_*)| \geq \zeta(2\sigma_*)/\zeta(\sigma_*)$. When

$\sigma < \frac12$, $|F(\sigma+iT')| = |F(1-\sigma+iT')|$ at $T' = \operatorname{Im} s_*$. The Gamma ratio from s_* to $1-\sigma+i\operatorname{Im} s_*$ contributes $\exp\big(\frac12\ell_*(1-\sigma-\sigma_*)+\vartheta(1-\sigma-\sigma_*)\big)$, which with $|e^{-\alpha(s_*)v}| = e^{\ell_*(\sigma_*-\sigma)/2}$ gives the stated exponent.

Gaussian integration by parts on $[-V,\infty)$, $\int_{-V}^{\infty} v\varphi_\tau\Theta = \tau^2\varphi_\tau(V)\Theta(-V) + \tau^2\int_{-V}^{\infty}\varphi_\tau\Theta'$, turns Lemma 3.3 into $\mathcal{N} = \varphi_\tau(V)\Theta(-V) + \mathcal{J} + \mathcal{N}_L$ with $\mathcal{J} = \int_{-V}^{\infty}\Delta\Theta\varphi_\tau$ and $\mathcal{N}_L = \int_{-\infty}^{-V}(v/\tau^2 - c_1)\Theta\varphi_\tau$. Applied once more to $\Delta_1\int_{-V}^{\infty} v\Theta\varphi_\tau$, it gives

$$(1-\tau^2\Delta_1)\mathcal{J} = \tau^2\Delta_1\big(\varphi_\tau(V)\Theta(-V) + c_1\mathcal{D}_E\big) + \int_{-V}^{\infty}\rho\Theta\varphi_\tau, \qquad \mathcal{D}_E = \int_{-V}^{\infty}\Theta\varphi_\tau.$$

Hence, with $\mathcal{D}_L = \int_{-\infty}^{-V}\Theta\varphi_\tau$ and $\mathcal{D} = \mathcal{D}_E + \mathcal{D}_L$,

$$\mathcal{N} - \frac{\tau^2\Delta_1 c_1}{1-\tau^2\Delta_1}\mathcal{D} = \frac{\varphi_\tau(V)\Theta(-V) + \int_{-V}^{\infty}\rho\Theta\varphi_\tau - \tau^2\Delta_1 c_1\mathcal{D}_L + (1-\tau^2\Delta_1)\mathcal{N}_L}{1-\tau^2\Delta_1},$$

whose modulus is at most d_0 times the second fraction of E, because $|c_1| \le \kappa_*$, $|\mathcal{D}_L| \le D_L$ and $|\mathcal{N}_L| \le N_L$. Finally

$$\mathcal{D} = e^{\tau^2 c_1^2/2} + \int_{-V}^{\infty} e^{c_1 v}\big(e^{\omega}-1\big)\varphi_\tau + \int_{-\infty}^{-V}\big(\Theta - e^{c_1 v}\big)\varphi_\tau$$

and $|e^{\tau^2 c_1^2/2}| \ge e^{-\tau^2\kappa_*^2/2}$, so $|\mathcal{D}| \ge d_0$. Divide. □

Theorem 4.6 (transport law and floor)**.** *Under the hypotheses of Proposition 4.5,*

$$\Big|S(x,p,t) - \tfrac12 \operatorname{Re}\tfrac{F'}{F}(s_*)\Big| \le \tfrac12 E, \qquad S(x,p,t) \;\ge\; \tfrac14\ell_* - \tfrac12\kappa(\sigma_*) - \tfrac12 E.$$

For the family every quantity in E is a function of $(\ell_*, \tau, p, \mathfrak{a}, V)$, so the bound at x is $\mathfrak{F}(\ell_*(x))$ with

$$\mathfrak{F}(\ell) = \tfrac14\ell - \tfrac12\kappa\big(\sigma_0 + \tfrac12\tau^2\ell\big) - \tfrac12 E(\ell).$$

For ζ, E depends on ℓ_* and, through T and through c_* in $\mathcal{Q}_*$, on the height; c_* may be bounded by its value at $\operatorname{Re} z = \frac12\sigma_*$, $|z| = \frac12(\sigma_*^2+T^2)^{1/2}$.

Proof. Lemma 3.1, Proposition 4.5, and $\frac12\operatorname{Re}(F'/F)(s_*) = \frac14\ell_* + \frac12\operatorname{Re} c_1 \ge \frac14\ell_* - \frac12\kappa_*$. In the family case $\mathcal{Q}_*$, $\mathcal{Z}$, ϑ, r, g and ω_m are functions of real arguments fixed by $\sigma_* = \sigma_0 + \frac12\tau^2\ell_*$ and ℓ_*. □

Remark 4.7 (the transport reading). To second order in τ the deviation is $\tau^2\big(\Delta_1 c_1 + \frac12\Delta''(0)\big) = \frac12\tau^2(\mathcal{L}''/\mathcal{L})'(s_*) + \tau^2\big(\alpha' c_1 + \frac12\alpha''\big)(s_*)$; the evaluation of Section 6.5 confirms this for ζ. This deviation is $O(\tau^2)$ uniformly in the height, while the shift from w to s_* moves the main term by about $\frac14\tau^2 K_2\ell_*$. So, apart from the deviation, the heat flow at height p is evaluation of the time-zero quantity at $s_* = w + \frac{t}{2}\alpha(s_*)$: at height $p + \frac{t}{2}\ell_*$ instead of p, and at real part $x + t\operatorname{Im}\alpha(s_*) \approx x + \frac{\pi t}{4}$. The second shift matches the drift of the zeros toward the origin at speed about $\pi/4$ that the zero dynamics $\dot z_j = 2\sum_{k\ne j}(z_j - z_k)^{-1}$ give in the mean-field approximation: the Hilbert transform of the density $\frac{1}{4\pi}\log\frac{|x|}{4\pi}$ contributes $-\pi/8$ to the sum at $x > 0$. This is a heuristic and is used in no proof.

Remark 4.8 (the four contributions). Compared with the time-zero floor, the floor of Theorem 4.6 gains $\mathfrak{g}(t) = \frac12\big(\kappa(\sigma_0) - \kappa(\sigma_*)\big)$, plus $\frac14\big(\psi(\frac{\sigma_*+\mathfrak{a}}{2}) - \psi(\frac{\sigma_0+\mathfrak{a}}{2})\big)$ for the family at $x = 0$, and loses $\frac12 E$. The loss collects four contributions: the curvature of $\log\mathcal{L}$, of order $\frac12\tau^2(K_2\kappa + \frac12 K_3)(\sigma_*)$; the curvature of $\log\Gamma_F$, of order $\frac12\tau^2(A_1\kappa + \frac12 A_2)(\sigma_*)$ with $A_2(\sigma) = -\frac18\psi''(\frac{\sigma+\mathfrak{a}}{2})$, which matters for the

family at small height; the Gaussian mass left of $\operatorname{Re} s = 1$, of order $\exp(-(\sigma_* - 1)^2/2\tau^2)$ times a power of $\mathcal{Q}_*$; and the reflected term, of order $e^{-p\ell_*/2}$. As $t \to 0$, with $\ell = 2\operatorname{Re}\alpha(w)$ the limit of ℓ_*, $\mathfrak{g}/\tau^2 \to \frac{1}{4}K_2(\sigma_0)\ell$ while the first two contributions divided by τ^2 tend to $\frac{1}{2}(k\kappa + \frac{1}{2}k_2)(\sigma_0)$ with $k = K_2 + A_1$ and $k_2 = K_3 + A_2$. So at small times the floor does not drop once $\ell > 2(k\kappa + \frac{1}{2}k_2)/K_2$ at σ_0. The strip term carries the Gaussian factor $\exp(-(\sigma_* - \frac{1}{2})^2/2\tau^2)$ against a growth of about $\mathcal{Q}_*^{1/2}$ at $\sigma = \frac{1}{2}$. Since $(\sigma_* - \frac{1}{2})^2/2\tau^2 = \frac{p^2}{8\tau^2} + \frac{p\ell_*}{4} + \frac{\tau^2\ell_*^2}{8}$, it is exponentially small in ℓ_* when $p > 2$; for $1 < p \le 2$ its smallness comes from the first and last of these terms.

Remark 4.9 (premises). Theorem 4.6 uses the Hermitian functional equation with a degree-one Gamma factor; the Euler product, only through Lemma 4.1 (the coefficients of $\log\mathcal{L}$ are dominated by those of $\log\zeta$); bounded coefficients, through Lemma 4.4; and about zeros only that H_t has none with $|\operatorname{Im} z| > 1$. Linear combinations of L-functions of Davenport–Heilbronn type satisfy the first and third premises and fail the second. The statement (2.3) is a smoothing statement at height $p > 1$; it constrains zeros near the axis only through a comparison argument such as Proposition 2.2, together with the confinement and first-contact inputs of Section 2.4.

5 A bound for $\partial_p S$

The comparison of Section 7 also needs an upper bound for $\partial_p S$. Cauchy's estimate gives one from Proposition 4.5.

Corollary 5.1 (derivative bound)**.** *Let $r > 0$. Suppose that at every point w' of the closed disk $|w' - w| \le r$ the hypotheses of Lemma 3.2 and Proposition 4.5 hold, with $E(w')$ the bound (4.2). For the family this includes $\operatorname{Re} w' > 1$ and $\log\frac{q}{\pi} + \psi(\frac{\operatorname{Re} w' + \mathfrak{a}}{2}) > 0$; for ζ it includes $\operatorname{Re} w' > 1$, $\operatorname{Im} w' \ge 20$ and $2\operatorname{Re} w' - 1 \le 2\operatorname{Im} w'$. Let $a_* \ge |\alpha'(s_*)|$ with $\tau^2 a_* < 1$, for instance $a_* = \frac{1}{4}\psi'(\frac{\sigma_* + \mathfrak{a}}{2})$ for the family and $a_* = 2/T$ for ζ (Lemma 4.2). Then*

$$\partial_p S(x,p,t) \;\le\; \frac{K_2(\sigma_*) + a_*}{4(1 - \tau^2 a_*)} + \frac{1}{4r}\max_{|w'-w|=r} E(w').$$

Proof. On the disk, $\mu \mapsto \tau^2\alpha(w' + \mu)$ maps a fixed domain into itself with a contraction constant uniform in w'. For the family the domain is the half-plane $\operatorname{Re}\mu \ge 0$. For ζ it is D_{x_+} of Lemma 3.2, with x_+ the largest value of $2\operatorname{Im} w'$ on the disk. The estimates in the proof of that lemma apply to $\mu \in D_{x_+}$, because there $|s| \le x' + 2 + \frac{1}{8}\log x_+ \le 2x'$ with $x' = 2\operatorname{Im} w'$, so the images lie in $D_{x'} \subseteq D_{x_+}$. The iterates from $\mu = 0$ are holomorphic in w' and converge uniformly, so $s_*(w')$ is holomorphic, and implicit differentiation gives $s_*' = 1/(1 - \tau^2\alpha'(s_*))$.

The Gaussian average $\mathcal{G}$ has no zero with $\operatorname{Re} w' > 1$ (Lemma 3.1). Hence $\varepsilon(w') = (\mathcal{G}'/\mathcal{G})(w') - (F'/F)(s_*(w'))$ is holomorphic on the disk, and $|\varepsilon| \le E$ there by Proposition 4.5. Cauchy's estimate gives $|\varepsilon'(w)| \le \max_{|w'-w|=r}|\varepsilon(w')|/r$. Since $S = \frac{1}{2}\operatorname{Re}(\mathcal{G}'/\mathcal{G})(w)$ and $\partial w/\partial p = \frac{1}{2}$,

$$\partial_p S = \tfrac{1}{4}\operatorname{Re}\big[(F'/F)'(s_*)\, s_*'(w) + \varepsilon'(w)\big].$$

Finally $|(F'/F)'(s_*)| \le |(\mathcal{L}'/\mathcal{L})'(s_*)| + |\alpha'(s_*)| \le K_2(\sigma_*) + a_*$ by Lemma 4.1 with $j = 1$, and $|s_*'| = |1 - \tau^2\alpha'(s_*)|^{-1} \le 1/(1 - \tau^2 a_*)$. □

Remark 5.2. The disk condition matters for ζ at small heights. There the pole of α at $s = 1$ gives s_* branch points with $\operatorname{Re} w$ slightly above 1, about $1.055 \pm 0.625i$ at $t = 0.2$ (found by a same-model check, see Appendix B). Every use below is at $x \ge X$. In the evaluation, the maximum over the circle is taken at its leftmost point, height $p - 2r$, and at the smallest ℓ_*; this rests on the sampled monotonicity of Section 6, so the evaluated bound is numerical.

6 Numerical evaluation

This section evaluates the bound E of Proposition 4.5 in floating point. Three things separate it from a statement for every t. First, E is evaluated without outward rounding. Second, the decrease of E in ℓ_*, which turns an evaluation at the smallest ℓ_* into a bound at every height, is sampled rather than proved. Third, the times form the grid

$$\mathcal{T} = \{0.001, 0.002, 0.005, 0.01, 0.02, 0.03, 0.05, 0.075, 0.1, 0.125,\\ 0.15, 0.175, 0.2, 0.225, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5\},$$

and below $t = 0.001$ only the leading order of Remark 4.8 is controlled. The programmes and records are listed in Appendix B; R1 is the record of the first programme, with sections A–H.

6.1 Method

ζ and its derivatives are computed by Euler–Maclaurin summation at real and complex points, and ψ, ψ', ψ'' by recurrence and asymptotic series; both are checked in R1, section A. The real-variable majorants are tabulated on logarithmic grids with relative interpolation error below $4 \cdot 10^{-5}$, the integrals of Proposition 4.5 use composite Simpson rules, $V = \sigma_* - 1 - \eta$ with the better of $\eta \in \{0.03, 0.08\}$, and the integrals are cut at 16τ to the right of σ_*, at 40τ to the left, and in the reflected region at $1 - \sigma = \sigma_* + \tau^2\ell_* + 30\tau + 2$. An independent implementation written by a same-model check (see Appendix B) differs from the programme by up to $4 \cdot 10^{-5}$ relative, in both directions.

6.2 Thresholds in ℓ_*

Let $\ell_1(p)$ be the smallest ℓ with $\mathfrak{g}(t) \geq \frac{1}{2}E$ at every $t \in \mathcal{T}$, counting only the κ part of the gain (Remark 4.8), found by bisection to 0.03 (R1, section H). Table 2 lists it with the small-time threshold $2(k\kappa + \frac{1}{2}k_2)/K_2$ of Remark 4.8 at $p = \sqrt{5}$. For ζ the Gamma terms use $T = 10^4$; they decrease in T, so the row is valid at every height $x \geq 2 \cdot 10^4 + 1$, where $\operatorname{Im} s_* \geq 10^4$.

Case	$p = \sqrt{5}$	$p = 2.5$	$p = 3$	$p = 4$	small-time threshold, $p = \sqrt{5}$
family, even	10.65	8.77	6.74	4.40	6.40
family, odd	10.12	8.22	6.19	3.92	5.99
ζ ($T = 10^4$)	9.62	7.67	5.56	3.29	5.67

Table 2: The threshold $\ell_1(p)$ (numerical, R1 section H).

Just above ℓ_1 the binding time is near $t = 0.2$: at $\ell = 10$ and $p = \sqrt{5}$ the even family fails at $t = 0.2$ by 0.09 (R1, section E, which uses an 11-point grid from $t = 0.002$). From $\ell = 11$ on, the margin is smallest at the smallest time of the grid, where it is τ^2 times the small-time coefficient. With the ψ part of the gain at $x = 0$, the small-time thresholds at $p = \sqrt{5}$ fall to 5.22 (even) and 5.38 (odd). E is nonincreasing in ℓ, for ℓ from 10 to 40, at all 12 sampled settings: the even family at $p = \sqrt{5}$, the odd family at $p = 3$ and ζ at $p = \sqrt{5}$, each at $t \in \{0.002, 0.05, 0.2, 0.5\}$ (R1, section H). This monotonicity is checked, not proved, and the other settings used below were not sampled. With the zeta row, (2.3) holds at every grid time from $x \approx 4\pi e^{9.62} \approx 1.9 \cdot 10^5$ at $p = \sqrt{5}$, and from $x = 2 \cdot 10^4 + 1$ at $p = 3$.

6.3 The family at every height

At $x = 0$ the saddle is real and $\ell_* = \ell_*^{(0)}$, and Lemma 3.2 gives $\ell_*(x) \geq \ell_*^{(0)}$ at every x. With $\mathfrak{F}$ increasing (sampled), $S(x,p,t) \geq \mathfrak{F}(\ell_*^{(0)}(t)) =: \mathfrak{F}_0(t)$ for every real x. Table 3 gives the smallest conductor for which $\mathfrak{F}_0(t) \geq C_{q,\mathfrak{a}}(p)$ at every $t \in \mathcal{T}$: the passing end of a bisection in $\log_{10} q$ to 0.006 (R1, section H), rounded up. These thresholds lie below those of Table 2 because the ψ part of the gain counts at $x = 0$.

Parity	$p = \sqrt{5}$	$p = 3$
even	$q \geq 1.084 \cdot 10^5$ $(\ell_0 = 9.50)$	$q \geq 1.32 \cdot 10^3$ $(\ell_0 = 5.46)$
odd	$q \geq 4.892 \cdot 10^4$ $(\ell_0 = 9.49)$	$q \geq 6.62 \cdot 10^2$ $(\ell_0 = 5.39)$

Table 3: Conductors above which $\mathfrak{F}_0(t) \geq C_{q,\mathfrak{a}}(p)$ on the grid (numerical).

At $q = 10^6, 10^8, 10^{10}, 10^{12}$ the excess $\min_{t\in\mathcal{T}}(\mathfrak{F}_0(t) - C_{q,\mathfrak{a}}(p))$ is positive for both parities and $p \in \{\sqrt{5}, 3\}$, from 0.0014 to 0.0077, attained at $t = 0.001$; at $t = \frac{1}{2}$ it is 0.37 to 0.97 (R1, section F). So on the grid (2.3) holds with $C(t) = C_{q,\mathfrak{a}}(p)$ at every real x, and the floor rises with t.

6.4 ζ beyond the verified height

For ζ the region of (2.3) is $x \geq X$. We take Polymath's height, rounded down to $X = 6 \cdot 10^{10}$ [15], and $X = 5 \cdot 10^{12} + 194858$, the height of the row of Polymath's Table 1 that Platt and Trudgian use for $\Lambda \leq 0.2$ [14], as recorded in [9], with $T = X/2 - 1$ and $\ell_X = \log\big((X/2-1)/2\pi\big) - 10^{-6}$, which is at most ℓ_* at every $x \geq X$ by Lemma 3.2. Table 4 gives the excess $\min_{t\in\mathcal{T}}\big(\mathfrak{F}(\ell) - (\frac{1}{4}\ell_X - \frac{1}{2}\kappa(\sigma_0))\big)$, with $\mathfrak{F}$ sampled at $\ell_X + \{0, 1, 3\}$ (R1, section F).

X	$p = \sqrt{5}$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
$6 \cdot 10^{10}$ $(\ell_X = 22.29)$	0.0051	0.0021	0.00088	0.00044	0.00025
$5 \cdot 10^{12} + 194858$ $(\ell_X = 26.71)$	0.0064	0.0026	0.0011	0.00054	0.00030

Table 4: Excess of the floor over $\frac{1}{4}\ell_X - \frac{1}{2}\kappa(\sigma_0)$ for ζ at $x \geq X$ (numerical).

All minima are attained at $t = 0.001$. They are thin at large p only because τ^2 is small there, so the floor at those times is close to $\frac{1}{4}\ell_X - \frac{1}{2}\kappa(\sigma_0)$. On the grid, then, (2.3) holds at every $x \geq X$ with the constant floor $\frac{1}{4}\ell_X - \frac{1}{2}\kappa(\sigma_0)$; the x-dependent floor $\frac{1}{4}\log\frac{x}{4\pi} - \frac{1}{2}\kappa(\sigma_0)$ is larger. The sampled heights include the band $3 \leq \operatorname{Im} z \leq 6$ on which [6] works.

6.5 Validation

The Gaussian representation (R1, section B). For ζ at $(x,p,t) = (3,\sqrt{5},0)$, $(10,\sqrt{5},0.1)$, $(20,3,0.3)$ and $(25,\sqrt{5},0.5)$, S from Polymath's kernel integral and S from the Gaussian average of ξ agree to $9 \cdot 10^{-16}$. The two routes share no formula beyond the definition of ξ.

The transport law (R1, sections C and D). For ζ at $x = 10^3$ and 10^4 ($\ell_* = 4.38$ and 6.68), $p \in \{\sqrt{5}, 3\}$ and $t \in \{0.02, 0.1, 0.3, 0.5\}$, we compare S from the Gaussian average of ξ with the main term $\frac{1}{2}\operatorname{Re}(F'/F)(s_*)$ and the second-order term of Remark 4.7. For characters (real characters modulo 10007, odd; 10009, even; 100003 and 1000003, odd; 1000033, even; and a cubic character modulo 10009) at $x \in \{0, 2, 10\}$, $p \in \{\sqrt{5}, 3\}$ and $t \in \{0.05, 0.2, 0.5\}$, S is computed from the kernel integral with Gauss-sum root numbers, and the main term as the kernel's S at time zero at $(2\operatorname{Im} s_*, 2\operatorname{Re} s_* - 1)$. The findings are as follows.

- The deviation $|\varepsilon|$ (for ζ, of $\mathcal{G}'/\mathcal{G}$; for characters, of its real part) is at most 0.044 for ζ and 0.117 for the characters. The largest is at $x = 0$ for the real character modulo 10009 at $t = \frac{1}{2}$.
- For ζ the second-order term accounts for most of the deviation: at $x = 10^3$, $p = \sqrt{5}$ and $t = 0.02$ it is 0.0047 against a measured 0.0048.
- $E \geq |\varepsilon|$ at all 124 points. The ratio $E/|\varepsilon|$ runs from 1.6 (even character modulo 1000033, $x = 0$, $p = 3$, $t = \frac{1}{2}$) to about 1300, with median 22. It is largest at $p = \sqrt{5}$ and moderate ℓ_*.
- $S(x, p, t)$ exceeds the time-zero floor at the same x at every point, including points below the thresholds of Table 2.

These points test the identity and the size of E at moderate heights. They do not test the regime of Section 6.4, where the theorem, not a computation, is the evidence.

6.6 Comparison with Hakimoglu-Brown's floor

Hakimoglu-Brown's Euler-product floors for S at height p [6, Propositions 8 and 11] have the same main term as Theorem 4.6, $\frac{1}{4}L - \frac{1}{2}\kappa(\frac{1+p}{2} + \frac{t}{4}L)$ with $L = \log \frac{x}{4\pi}$. Two identities explain this. His heat weight is the Gaussian average,

$$e^{t \log^2 n/4} = \mathbb{E}\, n^{-\tau N} \qquad (\tau^2 = t/2),$$

and his point $s_B = w_B + \frac{t}{2}\alpha_B(w_B)$, with $w_B = \frac{1}{2}(1 - iz) = \bar{w}$ and $\alpha_B(s) = \frac{1}{2s} + \frac{1}{s-1} + \frac{1}{2}\log\frac{s}{2\pi}$, is the complex conjugate of the first iterate of the saddle equation $s = w + \frac{t}{2}\alpha(s)$, with α in Stirling form; $\alpha - \alpha_B = -\frac{1}{6s^2} + O(s^{-4})$. He controls the remainder through Polymath's Riemann–Siegel coefficient bounds [15, Propositions 6.1–6.2] on the band $3 \leq \operatorname{Im} z \leq 6$ and a truncated, heat-weighted Dirichlet polynomial. Here it is controlled by the exact Gaussian representation and Euler-product majorants, at any $p > 1$.

Table 5 compares the two at his height $p = 61/20$ and his $X = 60000083951.5$, on intervals $[t^-, t^- + \frac{1}{200}]$ (R2, section B). His bounds are his expressions (31)–(32) with his own upper-bound inputs, which reproduce his certified Table 3 (R2, section A). Ours take the gain at t^- and E at t^+, with G from Corollary 5.1 and the best radius $r \in \{0.05, \ldots, 0.35\}$.

t^-	main term	his C^*	our C	his G^*	our G	main term of G
0	5.2974	5.2903	5.2958	0.2340	0.2174	0.2094
0.05	5.3867	5.3603	5.3777	0.1738	0.1601	0.1217
0.10	5.4408	5.4193	5.4316	0.1121	0.1111	0.0771
0.15	5.4761	5.4607	5.4683	0.0735	0.0769	0.0517
0.195	5.4982	5.4869	5.4917	0.0520	0.0562	0.0374

Table 5: Floors for S and bounds for $\partial_p S$ at $p = 61/20$, $x \geq X$ (floating; his values reproduce his certified ones).

The loss against the main term is 0.007–0.026 for his floor and 0.0016–0.0095 for ours, so our remainder is 1.7 to 4.4 times smaller. Our derivative bound is smaller for $t \leq 0.1$ and up to 8% larger after. At $p = 301/100$, against his Table 2 (Proposition 8, steps of $\frac{1}{100}$), our floor is larger by 0.012 to 0.025. These comparisons set a floating evaluation against a certified one: the advantage counts only once our evaluation is certified.

7 A check from the classical strip

This section restarts Hakimoglu-Brown's derivative-kernel comparison [6, Theorem 12] from the classical strip $M(0) \le 1$ instead of his hypothesis $M(0) \le \frac{9}{16}$. Its analytic inputs are those of [6], an unrefereed working draft. We use them as stated and do not re-derive them. Throughout, $X = 60000083951.5$, $b = \frac{1}{25}$ and $p = \frac{61}{20}$, so $B = p^2 = 9.3025$; the squared heights that [6] calls q_i are written Q_i.

7.1 Imported statements

The four statements below are quoted from [6], with its notation adapted to ours. They are not proved here.

Lemma 7.1 ([6, Lemma 2]; imported)**.** *For $0 \le t \le \frac{1}{5}$, a zero of H_t with $|\operatorname{Re} z| \le X$ satisfies $|\operatorname{Im} z| < \frac{1}{5}$.*

Its proof uses Polymath's boundary computation [15, Section 8.4, claim (a)], that $H_t(x+iy) \neq 0$ for $X \le x \le X+1$, $\frac{1}{5} \le y \le 1$ and $0 \le t \le \frac{1}{5}$; the verification of the Riemann hypothesis below height $\frac{1}{2}(X+1)$; backward Hermite splitting [15, Proposition 3.1(ii)]; and the classical strip theorem.

Lemma 7.2 ([6, Lemma 10]; imported)**.** *Let $p > 0$, $B = p^2$, $0 < A_2 < B$ and $\lambda, \mu \ge 0$. For $D \ge 0$ and $0 \le r \le A_2$ put $w = D + B - r$, $\Delta_{\mathrm{K}} = w^2 + 4Dr$ and*

$$\Pi(D,r) = 2\Delta_{\mathrm{K}}^2 - (D + A_2 + 3r)\Big(2\lambda w \Delta_{\mathrm{K}} - \mu(2w + 4B)\Delta_{\mathrm{K}} + 8\mu B w^2\Big). \tag{7.1}$$

If $\Pi \ge 0$ throughout that domain, then for $0 < h^2 \le A_2$, $0 \le v \le h$ and $(d,v) \neq (0,h)$,

$$K_h(d,v) \;\ge\; h\Big(\frac{\lambda}{p} K_p(d,v) - \mu\, \partial_p K_p(d,v)\Big).$$

For $\lambda = 1$ and $\mu = 0$ this is the kernel inequality in the proof of Proposition 2.2; the derivative term pays for $\lambda > 1$. We rechecked the proof of Lemma 7.2; it holds for every $A_2 < B$.

Proposition 7.3 ([6, Proposition 11]; imported)**.** *Let $p = \frac{61}{20}$ and $0 \le t^- \le t^+ \le \frac{1}{5}$. Put $\sigma_p = \frac{1+p}{2} = \frac{81}{40}$, $u = \sigma_p + 5.57\,t^-$, $u_8 = \sigma_p + 2.78\,t^-$, $\bar e = \frac{p-1}{2} - 2.78\,t^-$, $N_1 = 65536$, $N_* = 69098$, $L_* = 2\log N_* + 3$, and for $j = 0,1,2$*

$$\delta_j = \sum_{n=2}^{N_1} \big(e^{t^+ \log^2 n/4} - 1\big)(\log n)^j n^{-u} + \int_{N_1}^{\infty} (\log y)^j y^{-u_8}\,\mathrm{d}y,$$

$$\varrho_0 = 1.01\, N_*^{-p}\Big(\frac{N_*^{\bar e+1}}{\bar e + 1} + N_*^{\bar e}\Big), \qquad \varrho_1 = L_*\varrho_0, \qquad \varrho_2 = (L_*^2 + 10^{-9})\varrho_0,$$

$$\eta_0 = 10^{-6}, \quad \eta_1 = 4\cdot 10^{-5}, \quad \eta_2 = 1.6\cdot 10^{-3}, \quad \nu_1 = \tfrac{1}{2} + 10^{-10}, \quad \nu_2 = \tfrac{1}{4} + 10^{-10},$$

$$e_0 = \delta_0 + \varrho_0 + \eta_0, \qquad e_1 = \nu_1\delta_1 + \varrho_1 + \eta_1, \qquad e_2 = \nu_2\delta_2 + \varrho_2 + \eta_2 + 10^{-9},$$

$$\varepsilon_0 = \zeta(u)e_0, \quad \varepsilon_1 = \zeta(u)\big(e_1 + \nu_1\kappa(u)e_0\big), \quad \varepsilon_2 = \zeta(u)\big\{e_2 + 2\nu_1\kappa(u)e_1 + \nu_2\big(K_2(u) + \kappa(u)^2\big)e_0\big\}.$$

If $\varepsilon_0 < \frac{1}{2}$, then for $x \ge X$ and $t^- \le t \le t^+$

$$S(x,p,t) \;\ge\; C^* := 5.5716 - 10^{-6} - \nu_1\kappa(u) - \frac{\varepsilon_1}{1-\varepsilon_0},$$

$$\partial_p S(x,p,t) \;\le\; G^* := \nu_2 K_2(u) + \frac{\varepsilon_2}{1-\varepsilon_0} + \Big(\frac{\varepsilon_1}{1-\varepsilon_0}\Big)^2 + 2\cdot 10^{-10}.$$

These are [6, (28)–(32)], where $\zeta(u)$, $\kappa(u)$, $K_2(u)$, ν_j, ϱ_j and $\bar{e}$ are written Z, ψ, ψ_2, c_j, ρ_j and e. The proof uses [6, Lemma 3], an approximation of Riemann–Siegel type on $3 \le \operatorname{Im} z \le 6$ derived from Polymath's decomposition [15, (37)–(41)] and coefficient bounds [15, Propositions 6.1–6.2]; its Lemma 6; the normalizer bounds of its Section 3; its fixed-cutoff argument around (16); its estimate (27); and the elementary inequalities of its Section 8.2, among them $\frac{1}{4}\log\frac{x}{4\pi} > 5.5716$ and $\lfloor\sqrt{x/4\pi}\rfloor \ge N_*$ for $x \ge X$.

Theorem 7.4 ([6, Theorem 12]; imported)**.** *Assume $M(0) \le Q_0 < B = p^2$. Choose $0 = t_0 < \cdots < t_n \le \frac{1}{5}$, $d_i = t_{i+1} - t_i$, and $Q_0 > Q_1 > \cdots > Q_n \ge b$. For $0 \le i < n$ suppose that $C_i, G_i, \lambda_i, \mu_i \ge 0$ satisfy*

$$S(x,p,t) \ge C_i, \qquad \partial_p S(x,p,t) \le G_i \qquad (x \ge X,\ t_i \le t \le t_{i+1}),$$

that (7.1) with $(A_2, \lambda, \mu) = (Q_i, \lambda_i, \mu_i)$ is nonnegative for $D \ge 0$ and $0 \le r \le Q_i$, and that

$$k_i = 4\Big\{\frac{\lambda_i C_i}{p} - \mu_i G_i - \frac{2\lambda_i}{B - Q_i} - \frac{2\mu_i(B+Q_i)}{(B-Q_i)^2}\Big\} > 0.$$

If $e^{k_i d_i}(2 + k_i Q_{i+1}) \ge 2 + k_i Q_i$ for every i, then $M(t_n) \le Q_n$ and $\Lambda \le t_n + \frac{1}{2}Q_n$.

Its proof uses Lemmas 7.1 and 7.2, the zero dynamics of Proposition 2.2, the classical strip theorem, backward Hermite splitting, and [15, Theorem 1.5(i)], which places every nonreal zero at $|x| < \exp(C/t)$ with an absolute constant C, for compact attainment of a first contact.

7.2 Where the initial strip enters

Hakimoglu-Brown states that his hypothesis "fixes the initial value $q_0 = 9/16$ of the strip comparison in Theorem 12; the zero dynamics, the boundary input, the approximation on $3 \le y \le 6$, the kernel inequality, and all certificates are independent of it" [6, Section 1.1], and he calls Theorem 12 "a general conversion theorem". We checked this against the proofs. Lemma 7.1 uses the classical bound $|\operatorname{Im} z| \le 1$, not Q_0. The barrier argument of Theorem 7.4 needs the barrier to start above $M(0)$, together with the inputs listed after it. Lemma 7.2 needs only $A_2 < B$. With $Q_0 = 1$ we have $M(0) \le 1$, $Q_0 < B$, and a final time below $\frac{1}{5}$. His Section 1.1 lists other half-planes only for $\sigma < 1$, and his Remark 5 and Section 9 warn against extrapolating his closed one-step formulas to $\sigma = 1$. Those warnings concern only the closed formulas. Indeed, from $Q_0 = 1$ his closed formulas do not reach $\frac{1}{25}$ by $t = \frac{1}{5}$ [6, Remark 5], and neither does the comparison of Proposition 2.2 with the floors of Proposition 7.3 (R2, section C).

Proposition 7.5 (conditional on [6])**.** *Assume Lemmas 7.1 and 7.2, Proposition 7.3 and Theorem 7.4, together with the inputs their proofs use: from [6], Lemmas 3 and 6, the normalizer bounds of Section 3, the fixed-cutoff argument around (16), estimate (27) and the elementary inequalities of Section 8.2; from [15], the boundary computation of Section 8.4, equations (37)–(41), Propositions 3.1(ii), 6.1 and 6.2 and Theorem 1.5(i); the verification of the Riemann hypothesis to height $\frac{1}{2}(X+1)$ [13]; and the classical strip theorem. Then*

$$\Lambda \ \le\ \frac{18891}{100000} = 0.18891.$$

Proof. Apply Theorem 7.4 with $Q_0 = 1$, $n = 34$, $t_i = i/200$ for $i \le 33$, $d_i = \frac{1}{200}$ for $i < 33$ and $d_{33} = \frac{391}{100000}$, so that $t_{34} = \frac{16891}{100000}$, with $Q_{34} = b$ and the rows $(Q_i, C_i, G_i, \lambda_i, \mu_i)$ of Table 9 in Appendix A. The classical strip gives $M(0) \le 1 = Q_0 < B$. The certificate R3 (Appendix B) verifies the remaining hypotheses.

- In exact rational arithmetic: the chain ($t_{i+1} = t_i + d_i$, $Q_{i+1} < Q_i$, $Q_{i+1} \geq b$, $Q_{34} = b$, $t_{34} \leq \frac{1}{5}$; the last step $Q_{34} < Q_{33}$ is read off the table); the nonnegativity of (7.1) on every row, through the positivity of all $34 \cdot 8 \cdot 25 = 6800$ Bernstein coefficients of the compactification [6, (34)–(35)] on eight subintervals, the smallest being $1.4155 \cdot 10^{-3}$; $7.780 < k_i < 10.592$; every gate, with the exponential replaced by its Taylor sum $\mathrm{T}_{20}(y) = \sum_{j \leq 20} y^j/j! \leq e^y$, smallest margin $1.079 \cdot 10^{-5}$; and the two inequalities of [6, Section 8.2] displayed after Proposition 7.3. The other elementary inequalities of that section are taken from [6].

- In directed-rounding decimal arithmetic at 32 digits, with exp and log moved one unit outward and the other operations rounded toward the bound: on every row, $C_i \leq C^*$ and $G_i \geq G^*$ for Proposition 7.3 on $[t_i, t_i + \frac{1}{200}] \supseteq [t_i, t_{i+1}]$, with margins at least $2.05 \cdot 10^{-6}$ and $2.02 \cdot 10^{-6}$, and $\varepsilon_0 \leq 0.0191 < \frac{1}{2}$. Here $\zeta(u)$, $\kappa(u)$, $K_2(u)$ and the δ_j are replaced by the finite upper bounds of [6, Section 8.2]; C^* decreases and G^* and ε_0 increase in these inputs.

Theorem 7.4 then gives $M(t_{34}) \leq \frac{1}{25}$ and $\Lambda \leq t_{34} + \frac{1}{50} = \frac{18891}{100000}$. □

7.3 Control, construction and sensitivity

Control. The same programme, applied to [6, Table 3] ($Q_0 = \frac{9}{16}$), returns its printed certificates: the least Bernstein coefficient $623287907570432169922330885087/47926828249320250000000000000000000$ in row 21, a gate margin above $2.78 \cdot 10^{-8}$, scalar margins $2.05 \cdot 10^{-6}$ and $1.10 \cdot 10^{-6}$, $8.63 < k_i < 10.32$ and the final time $\frac{11963}{100000}$.

Construction (R2, section C). The C_i and G_i are Proposition 7.3 on $[t_i, t_i + \frac{1}{200}]$, rounded down and up to 10^{-4} with a margin of at least $2 \cdot 10^{-6}$; they do not depend on Q. Rows 0–23 coincide with [6, Table 3], except that G_6 is 10^{-4} larger here; his row 23 covers $[0.115, 0.11963]$ and ours $[0.115, 0.12]$. The λ_i, μ_i maximize k_i over a sampled admissible region, are reduced by 0.005, and were then rounded and lowered until the exact Bernstein test passed. Q_{i+1} is the barrier value $(Q_i + 2/k_i)e^{-k_i d_i} - 2/k_i$ rounded up to 10^{-6}, plus 10^{-6}.

Sensitivity (floating, R2 section C). A step of $\frac{1}{400}$ gives 0.18850, a step of $\frac{1}{100}$ gives 0.18961, and a polynomial margin of 0.02 gives 0.18982.

Thin places. The scalar margins of about $2 \cdot 10^{-6}$ are Hakimoglu-Brown's own; since rows 0–23 share his C_i and, except G_6, his G_i, a correction to his constants would affect both tables alike. In row 0 the leading coefficient of (7.1) in D, $2 - 2\lambda_0 + 2\mu_0 = 0.014$, is exact and positive. The smallest gate margin is $1.08 \cdot 10^{-5}$.

7.4 Literature

A same-model literature search, made after the table was built, read what it could fetch through a summarizing tool. Its quotations were not checked against the full texts, so Table 6 reports the claims as their authors state them.

No later version of [6] was found, and nothing stating 0.18891 or restarting its Theorem 12 from $Q_0 = 1$ turned up. Three imported inputs were checked. Platt [13] isolates the zeros of ζ with imaginary part up to 30,610,046,000, which exceeds $\frac{1}{2}(X+1) = 30{,}000{,}041{,}976.25$ by about $6 \cdot 10^8$. Polymath's Section 1 records the barrier region $X \leq x \leq X+1$, $y_0 \leq y \leq 1$, $0 \leq t \leq t_0$ with $t_0 = 0.2$, $X = 6 \times 10^{10} + 83952 - 0.5$ and $y_0 = 0.2$ [15]; its Section 8.4 could not be fetched in full. Polymath's Theorem 1.5 holds for $0 < t \leq \frac{1}{2}$ with an absolute, non-explicit constant, as the barrier argument of Theorem 7.4 needs.

Starting a stepped comparison from the classical strip is therefore not new, and Proposition 7.5 is not a record. Its distinctive feature is that its numerical inputs are published, refereed computations,

Source	Claim and status	Relation to Proposition 7.5
Gordon [5]	$\Lambda \le 0.157656619$, no half-plane hypothesis; "has not undergone external peer review"; uses Platt and Trudgian's verification to $3 \cdot 10^{12}$	dynamics of the highest zero with a reference barrier started at $(1000001/1000000)^2$, "the global strip gives strict entry at zero"; own analytic inputs and a far larger X
Raymond [16]	$\Lambda < 0.149396$; "has not undergone external peer review"	"starts from the classical width-one zero strip at time zero"; 608 time cells
Planat [11]	$\Lambda \le 0.148611789321711$ under stated premises	"a directed comparison beginning with the classical zero strip", using Raymond's source data; its appendix on [6] was not read
Gomila [4]	$\Lambda \le 0.1787854$, "not yet been peer reviewed"	Polymath's criterion at $X \approx 6 \cdot 10^{12}$ with a barrier computation; no strip comparison
Raymond, conditional [17]	$\Lambda < 0.116398$ assuming no zero with $\operatorname{Re} s > \frac{7}{8}$	below [6]'s 0.13963 under the same hypothesis

Table 6: Unrefereed 2026 claims found by the literature search.

Polymath's barrier and Platt's verification, with the analytic lemmas of [6] on top, checked by a 34-row certificate. The stronger claims rest on their authors' own unrefereed computations. For this paper the zeta value is a cross-check: with published numerical inputs only, a stepped comparison started from the classical strip already goes below the published 0.2, consistent with those claims.

8 Planning: what the floor would give

The values in this section are planning arithmetic. They combine the floating evaluation of Section 6 with comparison arguments, some of which are not yet written, and they bound nothing.

8.1 ζ with the transport floor

Replace Proposition 7.3 by Theorem 4.6 and Corollary 5.1, evaluated as in Section 6.6, and run Theorem 7.4, or the comparison of Proposition 2.2 ([6, Theorem 9]), at $X = 60000083951.5$ inside the window $t \le \frac{1}{5}$ of Lemma 7.1, with steps of $\frac{1}{200}$ (R2, section D). Table 7 gives the resulting planning values for Λ; a dash means that $\frac{1}{25}$ is not reached by $t = \frac{1}{5}$, or that $p^2 < 5Q_0$.

p	Thm. 7.4, $Q_0 = 1$	Thm. 7.4, $Q_0 = \frac{9}{16}$	Prop. 2.2, $Q_0 = 1$	Prop. 2.2, $Q_0 = \frac{9}{16}$
2.0	0.1953	0.1441	–	0.1491
2.25	0.1885	0.1382	0.2085	0.1524
2.5	0.1869	**0.1370**	0.2148	0.1561
2.75	**0.1867**	0.1374	–	0.1599
3.05	0.1882	0.1391	–	0.1643
3.5	0.1934	0.1424	–	0.1707

Table 7: Planning values for ζ with the floor of Theorem 4.6 (R2, section D).

So the transport floor would move 0.18891 and Hakimoglu-Brown's conditional 0.13963 by about 0.002–0.003. Heights below 2.5 do not help, for two reasons: the polynomial of Lemma 7.2 leaves less room for $\lambda > 1$ when $B = p^2$ is small, and K_2 grows as σ approaches 1, which raises G. Once certified, Theorem 4.6 and Corollary 5.1 would supply the inputs of such a table without

the Riemann–Siegel approximation of [6, Lemma 3]. The comparison itself would still rest on Lemmas 7.1 and 7.2 and Theorem 7.4.

8.2 The Dirichlet family

For Dirichlet L-functions the transport floor is the main new input. Take every primitive character of conductor q, start from $Q_0 = 1$, and let $b \to 0$: by Section 6.3, (2.3) then holds at every real x and no finite region is needed. Table 8 gives the time at which the barrier reaches 0, for the best p on a small grid. The first column uses the constant floor $C_{q,\mathfrak{a}}(p)$ with Proposition 2.2 [8, Section 4.6]; the second uses $\mathfrak{F}_0(t)$ with the same comparison (R1, section G); the third uses $\mathfrak{F}_0(t)$ and Corollary 5.1 with the kernel of Lemma 7.2 (R2, section E). R2's own run with the kernel of Proposition 2.2 agrees with the second column to about 0.001.

q	parity	constant floor	transport floor	transport floor, Lemma 7.2 kernel	change
10^6	even	0.3161 (2.85)	0.3076 (2.70)	0.2920 (4.0)	−7.6%
10^6	odd	0.3073 (2.69)	0.2990 (2.54)	0.2827 (3.5)	−8.0%
10^8	even	0.2618 (2.37)	0.2519 (2.37)	0.2344 (3.0)	−10.5%
10^8	odd	0.2541 (2.27)	0.2447 (2.27)	0.2268 (3.0)	−10.7%
10^{12}	even	0.1936 (2.24)	0.1868 (2.24)	0.1651 (2.5)	−14.7%
10^{12}	odd	0.1897 (2.24)	0.1839 (2.24)	0.1611 (2.5)	−15.1%

Table 8: Planning values for Λ_χ, uniform over primitive characters of conductor q, with the best p in parentheses. The last column compares the third column with the first, using the printed values.

The best p for the even character at $q = 10^6$ is at the edge of the grid. These values remain conditional on the uniform confinement of the nonreal zeros at positive times ([8], item H12), on a written comparison theorem for the family, now with Lemma 7.2's kernel and the second exterior input $\partial_p S \le G$, and on the certification of Section 6.

For a single character, Platt's verification heights [12] give $\ell \approx 16.6$ for every $q \le 400{,}000$ [8, Section 4.1], above every ℓ_1 of Table 2. So on the grid (2.3) holds with $C(t) = \frac{1}{4}\ell - \frac{1}{2}\kappa(\sigma_0)$ beyond the verified height of each such character. A bound for Λ_χ would still need a barrier for that character and a finite-region argument.

8.3 What remains

1. *Review.* Sections 2–5 should be reviewed first, since everything else rests on Theorem 4.6 and Corollary 5.1.

2. *Certification.* Re-evaluate E and the bound of Corollary 5.1 with outward rounding, with monotone majorants on intervals of σ and interval quadrature; cover every $t \in (0, \frac{1}{2}]$ by monotonicity or Lipschitz bounds between grid times; prove the decrease of E in ℓ_* term by term; and bound the $O(\tau^3)$ term below $t = 0.001$. If outward rounding moves $\ell_1(\sqrt{5})$ above 12, the conductor thresholds of Table 3 must be recomputed.

3. *The family.* Confinement (H12) and the comparison theorem [8].

4. ζ. An independent re-derivation of [6, Lemma 3 and Proposition 11], on which Proposition 7.5 rests.

5. *Literature.* Full readings of [5, 16, 11, 4].

A The table of Proposition 7.5

Here $p = \frac{61}{20}$, $t_i = i/200$, $d_i = \frac{1}{200}$ for $i < 33$, $d_{33} = \frac{391}{100000}$ and $Q_{34} = \frac{1}{25}$. Each next height is the following row's Q_i. The columns are $10^6 Q_i$, $10^4 C_i$, $10^4 G_i$, $10^3 \lambda_i$ and $10^3 \mu_i$; all entries are exact.

i	$10^6 Q_i$	$10^4 C_i$	$10^4 G_i$	$10^3 \lambda_i$	$10^3 \mu_i$	i	$10^6 Q_i$	$10^4 C_i$	$10^4 G_i$	$10^3 \lambda_i$	$10^3 \mu_i$
0	1000000	52902	2340	1474	481	17	356111	54036	1280	1804	1068
1	952036	52964	2322	1524	540	18	329670	54090	1225	1815	1096
2	904955	53031	2284	1574	609	19	304348	54143	1172	1824	1118
3	858867	53100	2234	1621	691	20	280116	54193	1122	1834	1139
4	813966	53172	2174	1644	738	21	256931	54242	1074	1844	1167
5	770525	53245	2107	1658	764	22	234774	54288	1028	1853	1188
6	728552	53318	2036	1671	788	23	213607	54334	985	1861	1208
7	688021	53390	1962	1684	815	24	193403	54377	944	1869	1227
8	648917	53462	1887	1698	841	25	174128	54419	905	1878	1255
9	611181	53533	1813	1710	865	26	155754	54459	867	1885	1273
10	574812	53602	1739	1723	892	27	138247	54498	832	1892	1291
11	539777	53670	1667	1735	916	28	121575	54536	798	1899	1308
12	506049	53736	1596	1748	944	29	105705	54572	766	1905	1325
13	473600	53800	1528	1759	967	30	90610	54606	736	1912	1341
14	442411	53862	1462	1771	995	31	76255	54640	707	1919	1367
15	412455	53922	1399	1782	1018	32	62616	54672	679	1925	1383
16	383698	53980	1338	1794	1046	33	49660	54703	653	1930	1397

Table 9: The 34 rows from $Q_0 = 1$ (certificate R3).

B Reproducibility and checks

B.1 Programmes and records

The programmes and records are in `papers/investigations-3/03_effective_outer_remainders/numerics/round-1/` of the repository [18]. They are standard-library Python and each prints one deterministic JSON record. Run each from that folder as

```
python3 -I <programme>.py > <RECORD>.json
```

P1 `height_p_transport_claude_20261017.py`, record R1 `HEIGHT_P_TRANSPORT_CLAUDE_20261017.json` (106 kB). Sections A–D are floating diagnostics, E, F and H a floating evaluation of Proposition 4.5, and G planning. A few minutes.

P2 `hakimoglu_brown_comparison_claude_20261017.py`, record R2 `HAKIMOGLU_BROWN_COMPARISON_CLAUDE_20261017.json` (19 kB). It imports P1. Section A reproduces printed values of [6]: its value (18), the gates of its Table 2, the margins of its Proposition 8, the margins of its Table 3 in floating point, and its Tables 6–7 at rows 0, 12 and 23. Section B is the floating comparison of Section 6.6; C builds the table of Appendix A, with exact Bernstein and gate checks and floating scalars; D and E are planning. A few minutes.

P3 `unconditional_strip_table_claude_20261017.py`, record R3 `UNCONDITIONAL_STRIP_TABLE_CLAUDE_20261017.json` (2 kB): the certificate used in the proof of Proposition 7.5, with the control on [6, Table 3]. A few minutes.

Their SHA-256 values are:

P1 `ab90cc7dee26ad872c0dff84e96a97bfea1d39838115fe6c277330f9a0b47690`
R1 `e1e2f2b2b19973006484b5f79aa3373a440d981ba46003e28762879fd290341a`
P2 `4f78eeab23aa25a676aa80eb61c24b4829d35789384497043c76c76f3139c3de`
R2 `460a3478b9f0f63af370e3ffb3e884e584f7e7d123cc1cd9e4ac3bf9a1ba660b`
P3 `94073d936706b5ca0ad6fa707a39e8566bee938c4cd4806a04af1122c5ef5935`
R3 `69997f80743411bb952c727f1e8c25b710fe28b14d8a2f995967fb08383eff67`

Each record regenerates byte for byte with Python 3.13.16 on x86_64 Linux, where it was generated. With Python 3.10.12 on aarch64 Linux, R3 replays byte for byte; R1 agrees in sections A and E–H and differs in 24 floats of sections B–D (three identity differences below 10^{-15} in size and 21 measured deviations, by at most $4 \cdot 10^{-11}$ relative); R2 differs in two floats of section A, at rounding level. A replay shows that the arithmetic is reproducible, not that the floating evaluation is outward-rounded.

B.2 Same-model checks

Two Claude subagents of the drafting session, with their own code, checked the notes this manuscript integrates [9, 10].

- For Sections 3–6 (mpmath): the identity of Lemma 3.1 at four points to 10^{-30}; the three forms of Lemma 3.3 to $4 \cdot 10^{-31}$; Lemmas 4.1–4.4 in about 23 000 randomized cases with no violation; the identity in the proof of Proposition 4.5 to $4 \cdot 10^{-42}$ for random entire Θ and to $8 \cdot 10^{-28}$ for the Θ of ζ; Binet's formula to 10^{-29}; and P1's E at five parameter points against its own implementation, with relative differences at most $4 \cdot 10^{-5}$, two of them with P1's value below its own.

- For Sections 5 and 7 (mpmath at 34 digits): that nothing in [6, Lemmas 2, 3 and 10, Proposition 11 and Theorem 12] uses the half-plane except through $M(0) \le Q_0$; every k_i and gate in exact rationals; the polynomial condition on all 34 rows with its own adaptive Bernstein scheme and a different compactification; C^* and G^* on all rows, with the same smallest margins; the comparison from $Q_0 = 1$, reaching $\frac{1}{25}$ at $t \approx 0.16865$; the identities of Section 6.6; and the formulas of Corollary 5.1, including the branch points of Remark 5.2.

Both checks found errors, corrected in the notes before this draft: status words stronger than a grid evaluation supports; a misquoted contraction constant; a missing invariant domain for the zeta saddle; the crude strip bound allowed where $\mathcal{Q}_*$ is not justified (numerically harmless); numbers that disagreed with the record; the bound $\beta' \le \frac{1}{12}$ of Lemma 4.3, at first only sampled; the domain of Corollary 5.1; an incomplete dependency list; the word "unconditional" for the table's conclusion; and Decimal powers in P3, now repeated directed multiplications.

Two further Claude subagents checked this manuscript against the notes, the records and [6], with their own code. They found no error in the proofs of Sections 2–5. Points they raised, corrected here, include: conductor thresholds of Table 3 that the notes had rounded below the passing values of the record; two numbers of Section 6.5 that the notes had interchanged; the claim that [6, Proposition 7] shares the main term of Theorem 4.6; the hypotheses of Corollary 5.1 on the disk; and the dependency list of Proposition 7.5. All these checks share the model family of the drafting model and are not independent review.

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